Stanislav Mircic

DOB: 16 Feb, 1982
Lives in Novi Sad, Serbia

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SUMMARY

Software developer with 20+ years experience in commercial programming. Past decade mainly in the area of Biomedical Engineering and Neuroscience but AI and Neural Networks have been my passion since high school days.

Since I have spent a big part of my career on R&D jobs I have developed a broad spectrum of skills, including development of cross-platform desktop apps (C++), mobile apps (iOS), embedded programming and electronics. I had a lead role in teams that developed dozens of software and hardware products.
For detailed portfolio please visit stanislavmircic.github.io

WORK EXPERIENCE

Backyard Brains — Mart 2014 — Present

Software Engineer

Embedded programming:
Developing firmware for numerous products and prototypes based on the TI MSP430, STM32, Atmel AVR microcontrollers, TI and Cypress Low Energy Bluetooth SOC and ESP32, ESP8266 WiFi SOC. Involved/consulted during design of hardware products.

C++ development:
Working on the Spike Recorder desktop cross-platform application in C++. Application reads stream of data from USB and audio card and applies numerous real time processing (FFT, IIR filtering, thresholding, spike detection etc.).

AI related:
- Mentoring over a dozen AI projects as part of the BYB Fellowship program based on tinyML, TensorFlow Lite, Edge Impulse, and the MATLAB Neural Network toolbox. All projects had neuroscience topics like EEG decoding, P300 surprise detection etc.
- Developing a support chat system in Python using OpenAI GPT-3/embeddings, GPT-4 and FastAPI.
- Real-time simulation of spiking neurons in microcontrollers, using Izhikevich, Hodgkin-Huxley, and hybrid models.
- NeuroRobot project: Designing and developing the hardware and software architecture of a robotic system that utilizes CNNs and Spiking Neural Networks to process sensor data and controls actuators in real time.

iOS related:
- Developing IOS applications in Objective C for iPhone, iPad and Apple Watch.
- Apple's MFi hardware and iAP2 lightning port devices development.

Mentoring over 40 Neuroscience related projects in the course six years of BYB Fellowship program.

UNIT — Jun 2009 — Present

Owner and Developer
- Research in Neuromorphic/Hebbian learning neural networks Python,
- PyTorch and PyBullet robot environment simulations in Python,
- iOS applications in Objective C,
- Web and desktop Adobe Flex/AIR,
- Web development JavaScript, JQuery

Various Employers — 2007 — Jun 2009

Freelance Developer

Adobe Flex/Air applications, web development, C# development.

Università degli Studi di Genova — Dec 2006 — May 2007

Researcher

Study visit and work on control software for Haptic robot for rehabilitation and research in human motorics based on RealTime Application Interface for Linux and Irrlicht 3D Graphical Engine. Neurolab, DIST, University of Genova.

EDUCATION

Master's degree, Electrical engineering — 2000 — 2006

Faculty of Technical Sciences, University of Novi Sad

Electrical Technician of Energetic section — 1996 — 2000

Technical High School, Sombor

LANGUAGE

English

Fluent speaking, writing, listening, reading.

AWARDS

Marie Curie Fellowship

Jan – May 2006

Marie Curie Fellowship for Researchers within Marie Curie Action Neurovers for a project Neuro-Cognitive Science and Information Technology Virtual University.

Best Young Researcher Paper in the field of biomedical technique

June 2006

For paper "Application of Neural Networks for Automatic Classification of Leukocytes" at 50th ETRAN Conference, Belgrade

SELECTED PAPERS

Neurorobotics Workshop for High School Students Promotes Competence and Confidence in Computational Neuroscience

Frontiers in Neurorobotics · Feb 13, 2020

https://www.frontiersin.org/articles/10.3389/fnbot.2020.00006/full

Grasshopper DCMD: An Undergraduate Electrophysiology Lab for Investigating Single-Unit Responses to Behaviorally-Relevant Stimuli

Journal of Undergraduate Neuroscience Education May 30, 2017 · May 30, 2017

http://www.funjournal.org/wp-content/uploads/2017/05/june-15-162.pdf

Correlation between Carotid Plaque Parameters and Frequency of Microembolic Signals

MD-Medical Data · Dec 1, 2011
https://www.md-medicaldata.com/files/05-MD%20Vol3%20No4%20Mircic.pdf

Application of Neural Network for Automatic Classification of Leukocytes 8th Seminar on Neural Network Applications in Electrical Engineering · Sep 27, 2006
https://ieeexplore.ieee.org/document/4147185

An Electrophysiological Investigation of Power-Amplification in the Ballistic Mantis Shrimp Punch
The Journal of Undergraduate Neuroscience Education (JUNE), Spring 2019, 17(1):T12-T19
https://pmc.ncbi.nlm.nih.gov/articles/PMC6650263/pdf/june-17-t12.pdf

Study while you sleep: using targeted memory reactivation as an independent research project for undergraduates. Adv Physiol Educ. 2025 Mar 1;49(1):1-10. doi: 10.1152/advan.00056.2024. Epub 2024 Oct 24. PMID: 39446136.
https://pubmed.ncbi.nlm.nih.gov/39446136/

Low-Cost Classroom and Laboratory Exercises for Investigating Both Wave and Event-Related Electroencephalogram Potentials. J Undergrad Neurosci Educ. 2024 Aug 31;22(3):A197-A206. doi: 10.59390/YNPH4485. PMID: 39355672; PMCID: PMC11441432.
https://pubmed.ncbi.nlm.nih.gov/39355672/